

CALCULUS 101 · RECITATION HOMEWORK

Homework 2: How Much Piles Up?

Name: _____

Date: _____

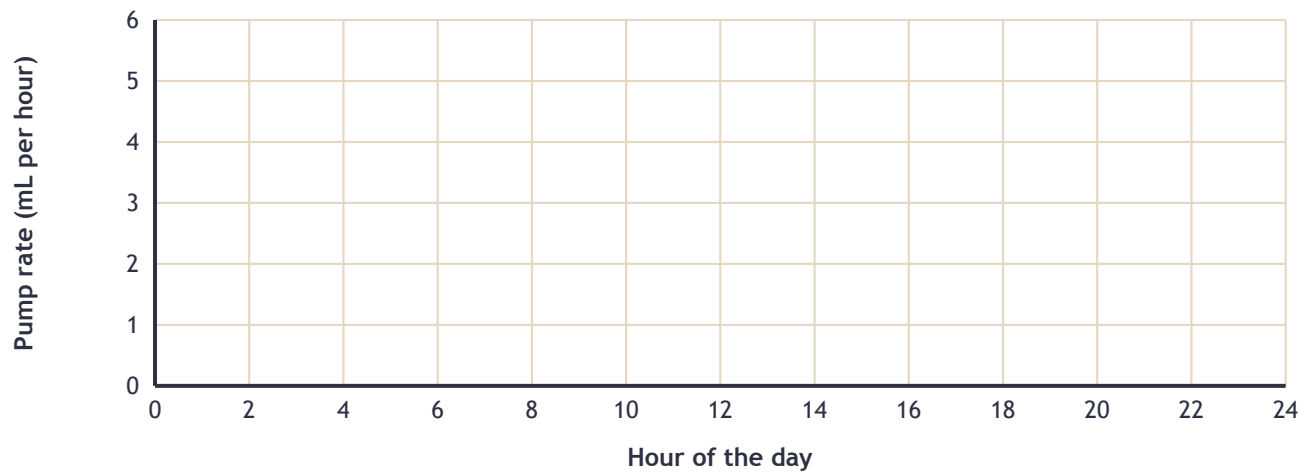
Instructions: About 20 minutes. Total = rate $\times$ time. Show your multiplication — the steps are worth points. Part B uses the grid: draw the staircase, then add the rectangles. Total: **21 points**.

Part A — Rate times time

1. Three pumps, three babies. Fill in the total each baby received. (Rate $\times$ hours. 2 points each: 1 for setting up the multiplication, 1 for the answer.) (6 pts)

Baby	Pump rate	Hours running	Multiplication	Total (mL)
A	3 mL/hr	24		
B	2.5 mL/hr	8		
C	5 mL/hr	12		

Part B — Draw the staircase



2. On the grid above, draw the pump's day: 2 mL/hr from hour 0 to 8, then 3 mL/hr from hour 8 to 16, then 4 mL/hr from hour 16 to 24. Shade under the steps. Then write the area of each rectangle and add them up for the day's total. (4 pts)

Part C — The In-and-Out sheet

3. A real-style I&O sheet. IN: the baby had 8 tube feeds of 18 mL each. OUT: four diapers weighed 32 g, 28 g, 35 g, and 30 g; a dry diaper weighs 20 g. Fill in the totals. (1 gram of wet weight = 1 mL of urine.) (4 pts)

	Work	Total (mL)
IN: feeds		
OUT: diaper 1 (32 g)		
OUT: diapers 2, 3, 4		
OUT: all four diapers		

Part D — Per kilogram

4. A baby weighs 1.2 kg and got 180 mL of milk today. What is that in mL/kg/day? Is she at 'full feeds' (about 150-160)? (2 pts)

Part E — Flashbacks

5. From NICU Tutor: what are trophic feeds for? (1 pt)

6. From last week's Calculus: a baby went from 1,000 g to 1,140 g in 7 days. Rate of change? (1 pt)

Part F — Think like a doctor

7. The pump was set to 3 mL/hr all day, and the I&O sheet says 72 mL. But you find out the pump was paused for 4 hours during a procedure and nobody wrote it down. What did the baby really get? Show it on a rate graph in words (what happened to the rectangle?), and explain why the difference matters to the team. (3 pts)

HOME LAB (optional, not graded): The dripping cup

You need: a paper or plastic cup, a pin or sharp pencil, a measuring cup with mL marks, a timer, a sink or tray

1. Poke a small hole in the bottom of the cup. Cover it with a finger, fill the cup with water, and hold it over the measuring cup.
2. Let go and time exactly 1 minute. Read the mL. That's your rate in mL per minute. Write it down.
3. PREDICT: at that rate, how much water will be in the measuring cup after 5 minutes? ($\text{Rate} \times 5$)
Write your prediction.
4. Empty the measuring cup, refill the drip cup, and run it for 5 minutes. Measure. Compare with your prediction.
5. Watch the drips near the end. Are they as fast as at the start?

What I saw:

Done! Professor's initials: _____

Score: _____ / 21 points

Professor's comment:
